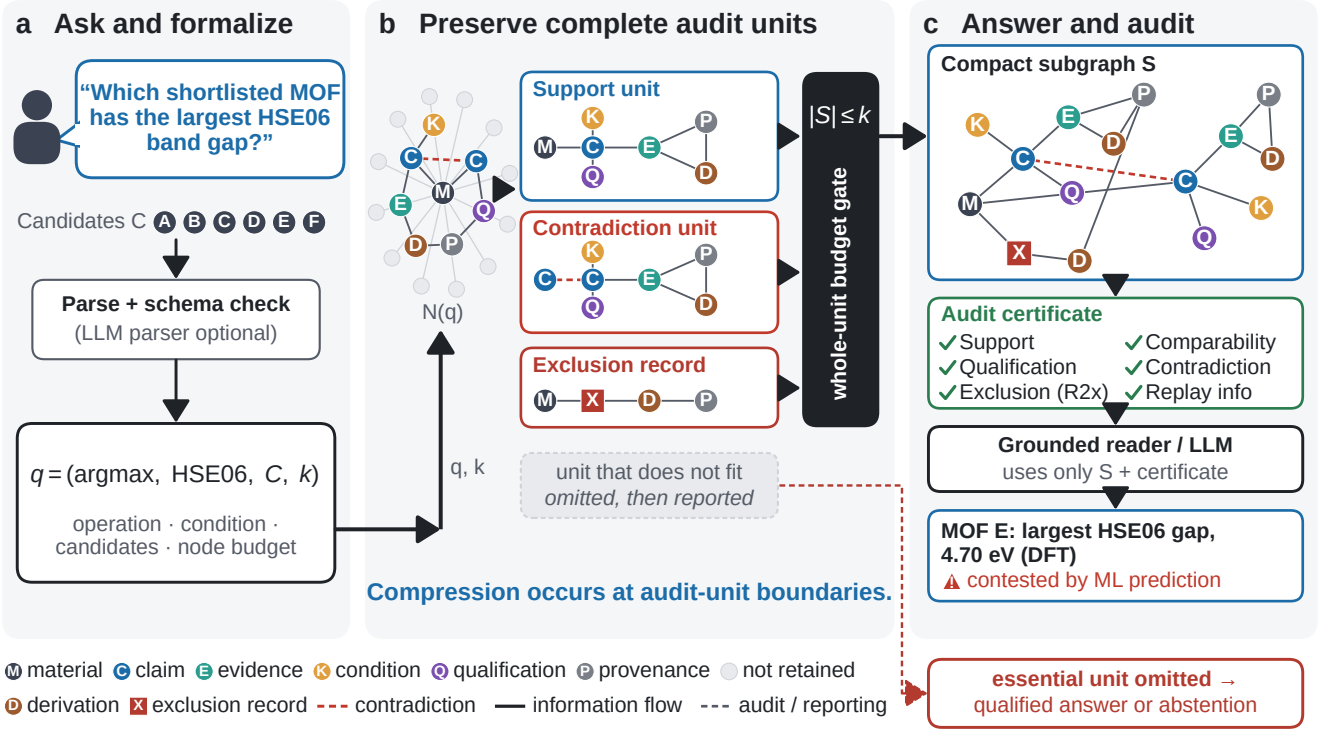

a Ask and formalize
"Which shortlisted MOF has the largest HSE06 band gap?"
Candidates C A B C D E F
Parse + schema check
(LLM parser optional)
q = (argmax, HSE06, C, k)
operation · condition · candidates · node budget
b Preserve complete audit units
N(q)
q, k
Support unit
Contradiction unit
Exclusion record
|S| ≤ k
whole-unit budget gate
unit that does not fit
omitted, then reported
Compression occurs at audit-unit boundaries.
c Answer and audit
Compact subgraph S
Audit certificate
Support
Comparability
Qualification
Contradiction
Exclusion (R2x)
Replay info
Grounded reader / LLM
uses only S + certificate
MOF E: largest HSE06 gap, 4.70 eV (DFT)
contested by ML prediction
material
claim
evidence
condition
qualification
provenance
not retained
derivation
exclusion record
contradiction
information flow
audit / reporting
essential unit omitted →
qualified answer or abstention